MIL-R-18401A(AS)
AMENDMENT - 7
31 May 1968
SUPERSEDING
AMENDMENT - 6
15 May 1963

MILITARY SPECIFICATION

RADIO SET AN/ARC-52 AND AN/ARC-52X

This amendment forms a part of Military Specification MIL-R-18401 (AER) dated 1 July 1959 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic MIL-R-18401(AER) is changed to MIL-R-18401A(AS) and throughout the specification Bureau of Aeronautics shall be changed to Naval Air Systems Command.

*Page 1, paragraph 1.1: Add

"(5) Communication by means of Radio Telephone when used in conjunction with Juliet-28 equipment (RT-332B/ARC-52 or RT-424A/ARC-52X and necessary controls."

*Page 1, paragraph 1.2: Add

"1c - Receiver-transmitter RT-332B/ARC-52

7-5/8	10-1/4	21-1/2	51	3.3.1

"1d - Receiver-transmitter RT-424A/ARC-52X

7-5/8	10-1/4	21-1/2	51	3.3.1"

*Page 2, paragraph 1.3: Add

"Juliet-28
Control C-8057/ARC

Page 2, paragraph 2.1, under SPECIFICATIONS Military: Add

"MIL-M-7793 Meter, Time Totalizing"

FSC 5821

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Page 3, paragraph 2.3: Add

"(4) Reordered Equipment - The following Specifications shall take precedence over those concerning performance requirements listed or referenced in 2.1 where equipment is reordered except in those areas where the contract specifies redesign or modification relating to performance.

MIL-E-5400B	24 May 1957
MIL-I-6181D	25 November 1959

Page 5, after paragraph 3.1.7: Add

"3.1.8 Dissimilar Metals - Aluminum may be used in contact with bronze (Oilite) bearing material.

"3.1.9 Extra Contacts in Receptacles - The requirement for unused extra connectors shall not be applicable to modular unit connectors for the Receiver-Transmitters RT-332/ARC-52 and RT-424/ARC-52X. The Control, Radio Set C-1607/ARC-52 and Control, Radio Set C-2027/ARC-52 may have a minimum of three extra unused contacts."

Page 5, after paragraph 3.2.3.1: Add

"3.2.3.2 Vibration - The requirement for equipment vibration with isolators removed shall not apply."

*Page 5, paragraph 3.2.4: Delete "56 pounds: and substitute: "58.0 pounds."

Page 5: Add

"3.2.6.3 Time Totalizing Meter- The equipment shall contain a time totalizing meter in accordance with Specification MIL-M-7793. The meter shall be included in RT-332/ARC-52, RT-332B/ARC-52, RT-424/ARC-52X, and RT-424A/ARC-52X and shall be connected to measure the total time filament power is applied to the Receiver-Transmitter. The time totalizing meter shall be located where it is readily available for reading without the necessity for removal of the pressurized cover."

*Page 7, paragraph 3.2.10(1), after "RT-332/ARC-52": Add

"and RT-332B/ARC-52"

paragraph 3.2.10(1) (a):

Delete "Transmit 390" and substitute: "Transmit 450"

paragraph 3.2.10(2), after "RT-424/ARC-52X": Add

"and RT-424A/ARC-52X"

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control in the fully clockwise (maximum gain) position.

Page 7, paragraph 3.2.14.1: Add

"except that reversible nameplates shall be used on the Receiver-Transmitters RT-332/ARC-52, RT-332B/ARC-52, RT-424/ARC-52X, and RT-424B/ARC-52X. One side of the nameplate shall include (AUTOMATIC RELAY) to identify the equipment when internally connected for this mode of operation."

*Page 8, paragraph 3.2.15: Delete and substitute:

"3.2.15 Interchangeability - The equipment shall meet the interchangeability requirements as specified in Specification MIL-E-5400.

"3.2.15.1 RT-332/ARC-52 and RT-424/ARC-52X - The Receiver-Transmitter RT-424/ARC-52X shall be the same in all respects as the RT-332/ARC-52, with exception of the 28 V DC power supply. All modules except those related to the respective power supplies and the external blower motor shall be interchangeable.

"3.2.15.2 RT-332B/ARC-52 and RT-424A/ARC-52X- The Receiver-Transmitter RT-424A/ARC-52X shall be the same in all respects as the RT-332B/ARC-52, with the exception of the 28 V DC power supply. All modules except those related to the respective power supplies and the external blower motor shall be interchangeable. Both shall be compatible with Juliet-28. The RT-332B/ARC-52 differs from the RT-332/ARC-52, and the RT-424A/ARC-52X from the RT-424/ARC-52X, respectively, in that they are modified to be compatible with Juliet-28.

Page 8, paragraph 3.2.17: Add

"However, antenna conducted interference due to the main receiver injection frequency at not more than half of the 18 total injection frequencies shall not exceed 300 microvolts."

*Page 8, paragraph 3.2.18.3: Add

"The auxiliary guard output of the RT-332B/ARC-52 and RT-424A/ARC-52X shall be terminated in a 600 ohm load, including means for measuring the audio output voltage."

*Page 9, paragraph 3.2.18.4: Under "Carbon Microphone Input" and "Dynamic Mike Input": Delete "10,000 cps" and substitute "20,000 cps".

*Page 9, paragraph 3.2.20.2: Add

"The sensitivity of the auxiliary audio and the auxiliary guard audio shall be such that when an RF input of 5 microvolts (open circuit)

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modulated 30% at 1000 cycles is applied, the signal plus noise-to-noise ratio shall be not less than 3 db for the auxiliary audio and not less than 6 db for the auxiliary guard audio.

Page 9, paragraph 3.2.20.2.1: Delete and substitute

"3.2.20.2.1 Degraded or Unusable Channels Due to Internal Interference Certain main receiver channel frequencies may be degraded or unusable due to internally generated interference. When the equipment is supplied with the guard frequency set for 243.0 mc, the main receiver sensitivity measurements shall be made with an input of 20 microvolts at frequencies of 241.1, 241.2 and 275.6 mc. Six unusable channels 242.1, 246.6, 253.3, 287.1, 322.8 and 339.0 mc are permitted which are independent of the guard receiver frequency."

*Page 10, paragraph 3.2.20.4: Add

"RT-332B/RT-424A Main Receiver
6 db 80 Kc min.
60 db 200 Kc max."

paragraph 3.2.20.9.1: Add

"The auxiliary guard output shall be 0.15 to 0.45 volt rms when loaded with 600 ohms."

Page 11, paragraph 3.2.20.9.3, second sentence: Delete and substitute

"When a 1000-microvolt signal modulated 30% at 1000 cps is applied, the output shall be at least 0.25 volt. When the modulation frequency is varied from 70 to 200 cps, the output for this frequency range shall be within +3.5 and -3 db of the output obtained at 1000 cps. When the modulation frequency is varied from 200 to 7000 cps, the output from this frequency range shall be within +2 and -3 db of the output obtained at 1000 cps."

*Page 11, paragraph 3.2.20.9.3: Add

"The RT-332B and RT-424A shall not vary more than +4 and -4 db from the output obtained at 1000 cps when the modulation frequency is varied from 70 to 20,000 cps."

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*Page 12, paragraph 3.2.20.13.1.1: Add

"Frequency, CPS	RT-332B/RT-424A Relative Output in db
150	-5 or lower
300	+4 to -4
600	+3 to -3
1000	0 (ref. level)
3000	+2 to -2
6000	+3 to -3
10,000	+4 to -4
15,000	+4 to -4
20,000	+4 to -4"

*Page 13, paragraph 3.2.20.17: Add

"Under the above receive conditions, the auxiliary audio and the auxiliary guard shall have a signal plus noise-to-noise ratio of at least 25 db."

*Page 14, paragraph 3.3.1: Delete and substitute

"3.3.1 Receiver-Transmitter Unit: RT-332/ARC-52, RT-332B/ARC-52, RT-424 /ARC-52X, and RT-424A/ARC-52X - The receiver-transmitter shall meet the following requirements.

*Page 15, paragraph 3.3.1.4: Delete "52.7 pounds" and add: "54.3 pounds."

*Page 17, paragraph 3.3.4: Delete and substitute:

"3.3.4 Automatic Relay Controls C-2791/ARC and C-2791A/ARC - These controls shall be in accordance with Specifications MIL-C-6781 and MIL-P-7788 and shall be suitable for installation in a console panel."

*Page 17, paragraph 3.3.4.1: Delete and substitute:

*3.3.4.1 Function - The functions of the units are as follows:"

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*Page 17: Add

*3.3.4.1.1 Function C-2791/ARC - The unit shall provide prime control of the AN/ARC-52 Radio Set and shall include all controls for operation and frequency selection of the RT-332/ARC-52, RT-424/ARC-52X, or RT-559/ARC Receiver-Transmitter unit when used with a C-1607/ARC-52 in an automatic relaying installation. During automatic relaying operation the headset audio is common for #1 and #2 equipments. Volume controls on the control units adjust the individual headset audio levels. The operator of either equipment can disable the automatic relaying operation and transmit by pressing the push-to-talk switch. It shall be possible to manually select any of the available frequencies and preset up to 19 channels on any of these available frequencies. The unit shall have red lighting within the plastic edge lighted plate in accordance with specification MIL-P-7788."

*Page 17: Add

"3.3.4.1.2 Function C-2791A/ARC - This unit shall function as outlined in paragraph 3.3.4.1.1 except as follows: It shall be compatible with Juliet-28 equipped aircraft. A special headset audio is provided to implement this function. The audio outputs are common for #1 and #2 equipments in relaying mode. The headset audio is disconnected from the UHF audio out when in secure relay mode to prevent a security breach."

Page 20, paragraph 4.3.1.3, after the fifth sentence following temperature, humidity and vibration conditions: Add

"The equipment may be vibrated in a rotary motion for the full 10 minutes."

NOTE: An asterisk preceding a page number denotes a change made to the specification which did not appear in the previous amendment.

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